Influence of Simulated Weightlessness on the Intramuscular and Oral Pharmacokinetics of Promethazine in 12 Human Volunteers

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The National Aeronautics and Space Administration (NASA) recommends using promethazine to prevent and treat space motion sickness, but pharmacologic responses in space and on Earth are different. Twelve volunteers were given 50 mg promethazine orally or intramuscularly before and after 48 hours of bed rest to simulate weightlessness. The maximum measured plasma concentration (C_{max}), time to C_{max} (t_{max}), and area under plasma concentration versus time curve from 0 to infinity (AUC_{inf}) were determined, and the bioequivalence was tested between bed-rest and ambulatory status for the intramuscular and oral routes as well as between both routes for bed-rest and ambulatory position. Simulated weightlessness did not influence the ratio $AUC_{bed\ rest}/AUC_{ambulatory}$ after intramuscular injection, whereas a significant increase (26%) in the ratio was seen after oral administration, probably because of a prolonged contact time between promethazine and the intestinal wall associated with an increase in the intestinal transit time. The AUC was 3-fold higher when the drug was administered by the intramuscular route during both positions. Thus, intramuscular administration could be a good alternative to the oral route.

Keywords: *Promethazine; pharmacokinetics; bed rest; weightlessness*

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Space motion sickness (SMS), during or after exposure to weightlessness in space flight, has existed throughout the history of the manned space programs and remains an operationally significant medical problem during short (5-10 days) flights of the space shuttle. Symptoms usually develop in the first 6 hours of the flight, peak at 24 to 48 hours, and are resolved after 72 to 96 hours.[1] Space motion sickness is characterized by nausea, vomiting, anorexia, flatulence, belching, hiccupping, yawning, sensitivity to sensory stimuli, mild apathy, impaired concentration, and subjective warmth. The relative absence of pallor can be explained by the effect of fluid shift on capillary circulation.[2]

The incidence of SMS is around 67% for a first flight, with symptoms categorized as mild (47%), moderate (38%), or severe (15%).[3] Since March 1989,[4] there has been a major reduction in the severity and duration of symptoms. This reduction may be due to the use of an intramuscular form of promethazine for treatment[4] and less use of transdermal and oral scopolamine/ephedrine. Graybiel and Lackner[5] evaluated intramuscular injections of several medications in subjects who experienced severe nausea and vomiting prior to the 30th Keplarian parabola and those who requested treatment. The authors found that 0.5 mg scopolamine intramuscularly and 50 mg promethazine intramuscularly provided relief of symptoms within 10 minutes in 72% and 78% of subjects, respectively. Similarly, Hordinsky et al[6] tested the efficacy of medications by applying a motion sickness stressor (rotating chair/head movements) twice a day in the course of a typical 7-hour oral (promethazine/ephedrine, 25 mg/25 mg) or

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intramuscular (promethazine, 25 mg) medication test day. Motion sickness tolerance was expressed as minutes to reach malaise III[7] while performing standardized head movements (repeated sequences of forward 90°, right 90°, backward 90°, left 90°, forward 90°) for 15 seconds and then sitting upright for 15 seconds until the next sequence of head movements began. Head movements were directed by taped commands while the chair accelerated clockwise at a constant $0.2/s^2$ to a maximum rotation of 240 deg/s, with an intermediate plateau of 10 minutes at 180 deg/s. Medications increased the resistance to motion sickness from 2 hours (significant) to 6 hours (significant) after drug administration for the oral promethazine/ephedrine and from 1 hour (not significant) to 5 hours (significant) after drug administration for the intramuscular promethazine.

As a result, scopolamine is no longer recommended as the first-line medication for the prophylaxis or treatment of SMS, and current National Aeronautics and Space Administration (NASA) medical operations policy recommends treatment of crew members with moderate to severe symptoms of SMS with intramuscular promethazine 25 to 50 mg.[1,8] This treatment is typically given in the presleep period.[1]

Although promethazine seems to be the best protective agent against motion sickness, it has led, particularly with the intramuscular administration, to a marked decrement in performance, extreme fatigue, and weariness during tests performed on Earth,[6] though at the present time these side effects have not been clearly demonstrated in flight. This lack of correlation between the data reported on Earth and in weightlessness may be due to an insufficient number of astronauts having received the intramuscular injection and/or to different pharmacokinetics of promethazine in flight. Given the small number of space flights and the difficulties of conducting experiments during missions, earthbound models for simulating weightlessness are used to study the physiologic, pharmacokinetic, and pharmacologic modifications observed during flights. The most used clinical model is the –6° head-down bed rest[9] (head-down tilt) that reproduces the bone and muscle losses and the fluid shift from the bottom to the top of the body that are observed in space crew members.[10,11] We selected this model first to determine the influence of simulated weightlessness on the pharmacokinetic parameters of promethazine administered by the oral and the intramuscular routes at a single dose of 50 mg and second to compare intramuscular and oral exposure of promethazine in ambulatory and weightless situations.

Table I Anthropometric Data for the 12 Volunteers

No.	Age (y)	Height (cm)	Weight (kg)
1	26	171.0	74.8
2	29	168.0	62.4
3	29	174.0	71.4
4	30	178.0	84.0
5	21	180.0	76.2
6	32	176.5	74.0
7	24	170.0	62.6
8	27	185.5	84.0
9	39	173.0	62.4
10	23	183.0	71.0
11	25	166.0	56.2
12	40	176.5	78.8
Mean	29	175.1	71.5
SD	6	5.9	9.0

SUBJECTS AND METHODS

Subjects

Twelve healthy nonsmoking male volunteers were selected for the study. Subjects were excluded if they had acute, chronic, or mental disease; a family history of gastrointestinal disorders; allergy to promethazine; drug or alcohol abuse; or abnormalities in standard laboratory tests. Local ethical committee (CCPPRB Toulouse I, Toulouse, France) approval was obtained prior to the start of the study in accordance with legal requirements, and all subjects gave their written consent. The details of the subjects are shown in Table I.

Study Protocol

The study was conducted in the Medes-IMPS (Space Clinic, Rangueil Hospital, Toulouse, France). Promethazine was given at a dose of 50 mg by the oral route (two 25-mg promethazine tablets) and by the intramuscular route by injection into the deltoid muscle. The intramuscular dose was verified by weighing the injection device before and after dosing.

Half of the subjects first received the intramuscular treatment, whereas the other group was dosed by the oral route. After a 1-month washout period, administration by the other route was conducted. In each group, administration was performed on day 1 (D1) in a sitting position. After administration on D1 and until day 3 (D3), volunteers could stand up and walk about: this period was considered as the ambulatory control. Then on day 4 (D4), subjects were submitted to –6° bed rest for 48 hours, and the same dosing was performed. This last period was considered as the bed-rest status.

Taking into account the plasma half-life of promethazine (10-12 hours[12]) and the time to maximum plasma concentration (t_{max}) after oral (1-3 hours[12]) and intramuscular (50 minutes-4 hours[13,14]) administrations, blood was sampled before and at 0.5, 1, 1.5, 2, 3, 4, 6, 8, 10, 12, 16, 24, 36, and 48 hours after dosing. Blood samples were collected into heparinized tubes (Vacutainer; Plymouth, United Kingdom), which were centrifuged and then the supernatants frozen at –80°C until assay.

Analytical Methods

Plasma promethazine assays were performed using a validated liquid chromatography method (HPLC) with UV detection. An analytical validation was performed according to the Arlington criteria[15] in the Pharmacokinetics and Clinical Toxicology Laboratory of the University Hospital of Rangueil, Toulouse, France, before starting and during the assays. A calibration curve ranging from 2 (lower limit of quantification, LLOQ) to 100 ng·mL^{-1} (upper limit of quantification, ULOQ) was constructed for the plasma assays and validated by 3 levels of quality controls. The LLOQ of the method was fixed at 2 ng/mL, taking into account the plasma half-life of promethazine (10-12 h[12]), the dosage (50 mg), the oral bioavailability (around 27% orally[12]), and the last sample time planned at 48 hours. Indeed after the oral administration of 50 mg, the plasma concentration assayed at 48 hours is assumed to be located around the LLOQ.[16]

Briefly, 1 mL plasma containing imipramine as internal standard was alkalinized and then extracted with 5 mL of a mixture of petrol ether and isoamylic alcohol (98/2 volume/volume). After mixing and centrifugation, the organic layer was evaporated under a nitrogen stream at 50°C. The dry residue was dissolved in 50 µL of the mobile phase (ammonium acetate buffer 50 mL, ammonia solution 28% 2 mL, and methanol sufficient quantity for 1 L) and injected onto a Nucleosil column (125 × 4.6 mm, 5.0-5.5 µm). Promethazine was detected at 254 nm.

Pharmacokinetic and Statistical Analysis

The interpretation of the drug time-concentration curves for each volunteer and for each administration route (intramuscular and oral routes) was performed with Kinetica software (InnaPhase, Champ-sur-Marne, France), using an independent-modeling calculation. We measured the following pharmacokinetic parameters: AUC_{0-t}, the area under the plasma concentration versus time curve from 0 to the last measurable concentration (C_{last}), calculated using the trapezoidal method; AUC_{inf}, the area under plasma concentration versus time curve from 0 to infinity, calculated as the sum of the $AUC_{0-t} + C_{last}/k_e$, where k_e is the first-order terminal constant rate calculated from a semi-log plot of a plasma concentration versus time curve; C_{max} and t_{max}, the maximum measured plasma concentration and the time to reach it, respectively; $t_{½}$, the terminal half-life calculated as $\ln(2)/k_e$. For the calculation of the half-lives, we only considered the data above the LLOQ (2 ng·mL^{-1}). The apparent clearance (Cl/F) was calculated as the ratio of the dose to the AUC_{inf}, and the apparent volume of distribution (Vd/F) was calculated as Cl/F divided by k_e. The bioequivalence was tested between the bed-rest and ambulatory positions ($AUC_{bed\ rest}/AUC_{ambulatory}$) for intramuscular and oral routes as well as between the intramuscular and oral routes (AUC_{IM}/AUC_{oral}) for bed-rest and ambulatory status.

All the pharmacokinetic parameters were expressed as arithmetic means except the ratio of AUC that are geometric means. Statistical analysis of the C_{max}, AUC_{inf}, k_e, $t_{1/2}$, Cl/F, and Vd/F was performed with an analysis of variance at one factor.[17] Statistical analysis of the t_{max} was performed with a Kruskal-Wallis test.[17] The bioequivalence was tested by the Schuirmann 1-sided *t* test. Statistical analysis was conducted using Kinetica software (InnaPhase). All the statistical tests were 2-sided except for the Schuirmann 1-sided *t* test and considered as significant for a *P* value less than .05.

RESULTS

Clinical Responses to Promethazine Administration

The promethazine was well tolerated in most of the subjects, but 3 of them suffered from cardiovascular (hypotension, bradycardia, or tachycardia) and/or neurologic (lipothymic sensation) adverse effects after the intramuscular injection in ambulatory conditions. The second intramuscular injection was not given to 1 subject (no. 11) and was given at half dose (25 mg) to 2 others (no. 2 and no. 10). To take into account the decrease in the dosage for these subjects, as promethazine presents linear pharmacokinetics for doses ranging from 25 to 50 mg,[18] plasma concentrations were considered to be twice the actual ones.

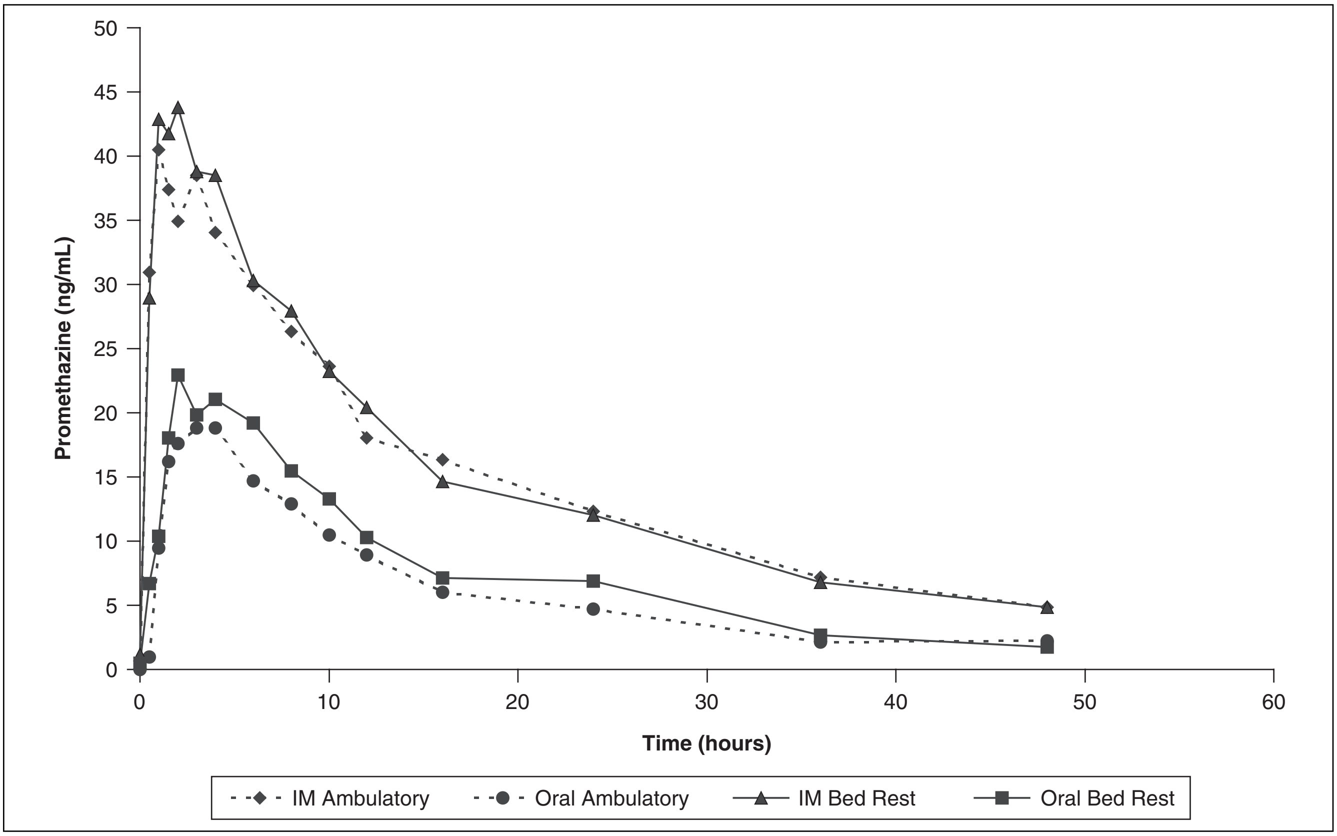

Figure 1. Mean of the plasma concentrations versus time measured after 50 mg promethazine administered by oral and intramuscular (IM) routes to 12 volunteers during ambulatory and head-down bed rest.

Pharmacokinetic Parameters of Promethazine

The pharmacokinetic profiles of promethazine versus time are shown in Figure 1. This graph shows that the plasma concentrations of promethazine are higher after intramuscular administration than after the oral route.

The corresponding pharmacokinetic parameters are shown in Table II. The statistical analysis (analysis of variance [ANOVA]: 1 factor) failed to show any difference between the 4 occasions (ambulatory intramuscular, ambulatory oral, bed rest intramuscular, bed rest oral) for k_e, $t_{½}$, and t_{max}, whereas the comparison showed statistically significant differences ($P < .05$) for C_{max}, AUC_{inf}, Cl/F, and Vd/F.

The bioequivalence was tested between bed rest and ambulatory position ($AUC_{bed\ rest}/AUC_{ambulatory}$) for intramuscular and oral routes as well as between the intramuscular and oral routes (AUC_{IM}/AUC_{oral}) for bed rest and ambulatory status. For the comparison between best rest and ambulatory status, the ratio was 104% ± 18% (not significant) and 126% ± 56% ($P < .05$) after intramuscular and oral administration, respectively. For the comparison between intramuscular and oral routes, the ratio was 345% ± 279% ($P < .05$) and 268% ± 154% ($P < .05$) for ambulatory and bed-rest positions, respectively. After intramuscular administration, the bioequivalence was demonstrated by the two 1-sided *t* test between bed-rest and ambulatory status. This finding showed that simulated weightlessness did not influence promethazine absorption after intramuscular injection into the deltoid muscle. This was not the case for the oral route. The AUC was significantly greater (26%) during simulated weightlessness than during ambulatory status. When intramuscular injection was compared with oral route, the AUC was around 3-fold higher, and these results are superimposable for the bed-rest and ambulatory situations.

DISCUSSION

The main aim of our study was to determine the influence of simulated weightlessness (–6° head-down bed

Table II Means (±SD) of the Pharmacokinetic Parameters Obtained After Intramuscular and Oral Administration of Promethazine to 12 Volunteers During Ambulatory and Bed-Rest Periods

		k_e h^{-1}	$t_{½}$ h	C_{max} µg·L^{-1}	t_{max} h	AUC_{inf} µg·L^{-1}·h	Cl/F L·h^{-1}	Vd/F L
Ambulatory	M	0.043	17.8	52.4	2.0	958	56.5	1364
Intramuscular	SD	0.014	5.7	15.9	1.1	299	16.1	329
Ambulatory	M	0.048	15.6	22.5	2.9	363	261.7	4848
Oral	SD	0.014	4.9	14.2	1.6	297	242.3	3243
Bed rest	M	0.043	19.3	55.4	1.6	1015	55.9	1386
Intramuscular	SD	0.017	10.2	11.6	1.0	457	17.6	428
Bed rest	M	0.055	15.3	31.2	3.2	438	188.6	3317
Oral	SD	0.027	6.7	23.0	1.9	362	129.6	1433
ANOVA (1 factor)		NS	NS	$P < .05$	NS	$P < .05$	$P < .05$	$P < .05$

k_e, first-order terminal constant rate; $t_{½}$, half-life; C_{max}, maximum observed plasma concentration; t_{max}, time to C_{max}; AUC_{inf}, area under the concentration-time curve from 0 to infinity; Cl/F, apparent clearance; Vd/F, apparent volume of distribution; ANOVA, analysis of variance; NS, not significant difference between the 4 occasions (ambulatory intramuscular, ambulatory oral, bed-rest intramuscular, and bed rest oral). $P < .05$: significant difference between the 4 occasions (ambulatory intramuscular, ambulatory oral, bed-rest intramuscular, and bed rest oral).

rest) on the pharmacokinetic parameters of intramuscularly and orally administrated promethazine. No significant difference was observed for k_e and $t_{½}$. These parameters depend on the volume of distribution (Vd) and the total clearance (Cl). Consequently, their fluctuations reflect the simultaneous variations of the Vd and the Cl, although these parameters are weighted by the bioavailability fraction (F) in the present study. The lack of significant differences for k_e and $t_{½}$ between the 4 occasions supposes that Vd and Cl did not change or changed in the same way. Promethazine undergoes an extensive first-pass hepatic metabolism ($E_H > 0.7$) with a Cl[19,20] very close to the hepatic blood flow. In a previous study performed during a 4-day head-down tilt,[21] we observed variations in the Cl of lidocaine used as a probe of hepatic blood flow.[22-24] Lidocaine was administered intravenously (7.1 mg/kg of body weight) on days 1 to 5. Blood samples were collected before and at regular intervals up to 360 minutes after drug administration, and the plasma concentrations were measured using a homogenous enzyme immunoassay (TDx, Abbott Diagnostics, Abbott Park, Ill). The hepatic clearance significantly increased between the ambulatory period and the first and the fourth days of head-down tilt, respectively, suggesting hepatic blood flow variations. Concerning the Vd, there is probably a pharmacokinetic impact of weightlessness because the Vd depends on the central volume (blood compartment), the peripheral volume (tissues), and the free fraction of the drug, and these parameters are sensitive to the body water fluctuations.[16] During true and simulated weightlessness, the loss of the hydrostatic pressure induces a redistribution of the organism's fluids (principally the venous blood) from the bottom to the top of the body.[25,26] This phenomenon is called fluid shift.[27] This fluid shift induces an increase in central venous pressure and implies an early adaptive response of the cardiovascular and renal/endocrine systems.[27,28] The physiologic mechanisms of adaptation reduce total body water and the plasma volume with reductions in the volumes of extracellular, intracellular, and interstitial fluids. These changes are reflected in a 2- to 4-kg decrease in body mass during the first hours of the space flight.[27,29,30] The hypovolemia due to the fluid loss stimulates a strong secretion of aldosterone and antidiuretic hormone. Similarly, an increase in sympathetic nerve activity was reported in flight compared with preflight data[31] most likely related to the concomitant decrease in plasma volume. A return to a normal state happens on the fourth day of space flight, characterized by stabilization of the body weight and of the fluid volumes of the astronauts until the end of the space flight.

The statistical analysis of the C_{max}, AUC_{inf}, Cl/F, and Vd/F showed significant differences (ANOVA, $P < .05$; Table II). All these pharmacokinetic parameters show a modification of the bioavailability. Indeed, bioavailability encompasses 2 aspects: first, the absorption rate characterized by t_{max} and C_{max} and then the absorbed fraction (bioavailability fraction F) characterized by C_{max} and AUC_{inf}.[32] Moreover, the lack of bioequivalence between bed-rest and ambulatory status after oral administration of promethazine clearly shows that simulated weightlessness acts on the mechanisms implicated in the oral absorption of promethazine. This is in agreement with the results that we obtained in a recent study performed with human volunteers who spent 3 months with a –6° head-down tilt and using acetaminophen as a gastric emptying probe.[33] Gastric emptying was assessed before –6° head-down bed rest for baseline data collection in a sitting position for the first 6 hours, after which time volunteers could stand up and walk about. During head-down bed rest, gastric emptying was studied 3 times: after 1 day, 18 days, and 80 days. After an overnight fast, the subjects were given 1 g acetaminophen, washed down with 200 mL water. Because the absorption of acetaminophen is used as a marker of gastric emptying[34-39] and because the rate of gastric emptying is altered by posture,[40-43] during the head-down tilt period measurements, the volunteers stayed in the head-down position, lying on their backs for the first 6 hours after administration, after which time they were permitted to lie flat on their stomachs but always remaining supine. Four hours after acetaminophen administration, the subjects were permitted to imbibe liquids, and after 6 hours, a standard meal was consumed. Blood and saliva were sampled before administration and at regular intervals up to 12 hours after dosing. For both plasma and saliva, we measured C_{max}, t_{max}, AUCs, and $t_{½}$. The plasma C_{max} increased and t_{max} decreased, evidence of more rapid drug absorption,[34-39] which increased with the duration of bed rest. All the plasma AUCs of acetaminophen were in the same order of magnitude, revealing that the total amounts absorbed were similar for all the bed-rest periods. We concluded that simulated weightlessness induced a more rapid absorption rate of acetaminophen, which might be explained by a faster gastric emptying but more likely by a more rapid intestinal absorption. This latter hypothesis could be related to cardiovascular variations.[44-50] A lower total peripheral resistance has been observed after 10 minutes in the supine position compared with the sitting[46] and the standing positions,[45-47] and after 6 hours in the –5° head-down tilt compared with the standing position.[51] When the total peripheral resistance decreases, the blood flow increases. This is also the case for the intestinal blood flow. Acetaminophen is totally (F = 80%-90%)[52] and rapidly absorbed (t_{max} = 30-90 minutes)[34] through the intestinal wall by passive diffusion.[53] For this kind of drug, the rate of diffusion depends directly on the gradient of concentrations between the intestinal lumen and the intestinal blood.[16] Usually, the intestinal luminal concentration is higher than the intestinal blood concentration because the intestinal blood flow maintains this difference. If the intestinal blood flow increases, the intestinal blood concentration of acetaminophen will decrease more rapidly and thus accentuating the gradient of concentrations between the lumen and the blood, resulting in an increase in the absorption rate of acetaminophen. The decrease in the total peripheral resistance observed in the supine position and in the head-down bed rest could be a physiologic explanation of the increase in the absorption rate as well as of the decrease in the t_{max} observed in our study. However, it should be noted that the duration of these experiments describing the decrease in the total peripheral resistance in supine and head-down positions was not very long (a few minutes to a few hours). This was not the case in our study in which the subjects were in the head-down position for much longer (80 days).

In the present study, bed rest did not influence the oral absorption rate because there was not significant variation of t_{max} but induced an increase in the bioavailability fraction. Different parameters may control the bioavailability fraction of a drug administered orally. The main ones are the intestinal transit time, the contact with luminal enzymes, the diffusion through the mucus and the intestinal wall, and the contact with intestinal and hepatic cytochromes.[54-56]

In space flight, as well as in models simulating weightlessness, intestinal transit is believed to be affected.[57] The intestinal transit time plays an important role controlling the bioavailability of the drugs that show slow absorption across the intestinal wall.[32] A longer intestinal transit time increases the contact time between the drug and the intestinal wall and promotes the amount of drug absorbed. On the contrary, a decrease in the transit intestinal time induces a decrease in the oral bioavailability fraction of the drug. At the present time, it is not clear if actual and simulated weightlessness significantly modify the intestinal transit time because no data are available. Consequently, we selected clinical studies that evaluated the effect of posture on intestinal transit time to indirectly estimate the influence of simulated weightlessness on transit time. At the present time, a number of techniques exist to determine the entire or partial intestinal transit time (pH telemetry, hydrogen breath test, scintigraphy, biomagnetic measurement system, radiography, nonabsorbable sulphurhexafluoride gas, etc.),[58-64] but there are only a few reports on the influence of posture on the transit time. Dainese et al[62] determined the effect of body posture, upright versus supine, on intestinal transit using nonabsorbable sulphurhexafluoride gas in 8 healthy subjects without gastrointestinal symptoms. In each subject, a gas mixture was continuously infused into the jejunum (12 mL/min) for 3 hours, and gas evacuation, clearance of nonabsorbable sulphurhexafluoride marker, and abdominal girth were measured. Paired studies were randomly performed in each subject on separate days in the upright and supine positions. In the upright position, intestinal gas retention was much lower than when supine (13 mL vs 146 mL retention at 60 minutes, respectively; $P < .05$), and clearance of sulphurhexafluoride marker was expedited (72% vs 49% at 60 minutes, respectively; $P < .05$). The gas challenge test was well tolerated both in the upright and supine positions without abdominal distension. The authors concluded that body posture has a significant influence on intestinal gas propulsion and that transit is faster in the upright position than when supine.

Before crossing the intestinal wall (the mechanism of the intestinal absorption of promethazine remains unclear, and it is not known if a transporter is implicated), promethazine is strongly bound to mucin[65] in the mucus layer that covers and protects the gastrointestinal mucosa.[32] When reaching the liver, promethazine undergoes an extensive first-pass hepatic metabolism ($E_H > 0.7$) with a total clearance[19,20] very close to the hepatic blood flow. At the present time, no data are available concerning the influence of simulated weightlessness on the different steps implicated in the intestinal absorption of a drug, whereas some data exist concerning the variations of the hepatic clearance. As described in a previous study,[21] after the first day of bed rest, an increase was observed in the hepatic clearance of lidocaine that undergoes an extensive first-pass hepatic metabolism ($E_H > 0.7$), as does promethazine. This increase in metabolism leads to a decrease in the bioavailability fraction and consequently in AUC, but that is not the case in the present study.

Taking into account the present results and the information concerning the influence of weightlessness on oral absorption, bed rest induced a higher bioavailability fraction of promethazine, probably related to a longer contact time between the drug and the intestinal wall due to a slower intestinal transit in the bed-rest position, even though the hepatic clearance may also be increased. The increase in the oral bioavailability of promethazine might indicate that products that have a similar oral absorption may attain toxic levels, especially if they have a narrow therapeutic index because such drugs present a small difference between therapeutic and toxic concentrations. Even if at the moment, no drugs with a narrow therapeutic index are present in the onboard medical kit,[66] these may be used in the future when many more persons will be participating in space travel and exploration.

It is interesting that the drowsiness and sedation reported after intramuscular administration of promethazine were subjectively much less evident in orbit than those resulting from a similar dose taken on the ground during a test before space flight.[67] Indeed, ground-based studies have shown the incidence of drowsiness associated with promethazine use to be 60% to 73%.[68,69] There has not been a general problem of this type reported in space shuttle flights. In fact, only 1 of the 21 crew members studied reported any symptoms relating to sedation, to give an incidence of 4.8%, which is significantly different from ground-based experience.[67] It might be considered that this lower sedative effect was due to a decrease in the intramuscular bioavailability of promethazine in weightlessness. That was not the case in the present study, and in fact, the difference is most probably related to the significant increase in catecholamines (epinephrine and norepinephrine) observed in flight[70,71] and not to a pharmacokinetic variation of promethazine.

In the second part of the study, we determined the ratio AUC_{IM}/AUC_{oral} of promethazine in ambulatory and simulated weightless situations. The AUC of promethazine was around 3-fold higher for the intramuscular administration compared with the oral, during ambulatory as well as bed-rest status. These results are in agreement with those reported by Schwinghammer et al.[72] These authors compared the C_{max}, t_{max}, and $AUC_{0\text{-}24h}$ of promethazine (50 mg) administered by the oral, rectal, and intramuscular routes to 24 healthy male volunteers. The mean $AUC_{0\text{-}24h}$ of the intramuscular injection was significantly higher (3-fold) than the values obtained for the oral and rectal routes, respectively. After oral administration, promethazine undergoes an extensive first-pass hepatic metabolism, resulting in a low bioavailability fraction, estimated to be around 25%.[19,20,73] Although the oral route is the easiest method of drug administration, its use is not always feasible. For example, for astronauts suffering vomiting, it is impossible to administer a drug orally because the amount absorbed can not be precisely quantified, and consequently, the situation justifies repeated administration with possible toxic concentrations. Therefore, another route of administration becomes necessary. The same situation may apply to hospitalized patients who maintain a bed-rest position for some days or weeks and who present with nausea and vomiting treated by using promethazine.

In conclusion, we observed that simulated weightlessness induced a significant increase in the oral bioavailability of promethazine, which was probably related to a longer contact time between the drug and the intestinal wall due to a slower intestinal transit in the bed-rest position. The lower bioavailability of promethazine when administered orally compared with intramuscularly is due to the extensive hepatic first-past metabolism. Added to this higher bioavailability, which requires the dose to be decreased when promethazine is administered by the intramuscular route, this route has a number of advantages, particularly in the case of astronauts suffering from vomiting. But all these observations will need to be validated by measuring salivary promethazine[33] during actual spaceflights.

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